

Direct evaluation of the M-ENDPOINT constant and the corrected sunset-axis endpoint sign

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

first issued 2026-06-07 · this version issued 2026-06-07 · v1.0

1. Purpose and scope

The SC-SCOPE all-orders lift is gated by four registered inputs (M-ENDPOINT, GHAT3-Q0 optional, GHAT4-PERTRANSFER, R-U6-1). This note resolves the first: M-ENDPOINT, the convexity-honest dressing variance $M(\hat{r}(2 \times 10^{-3})) = M(0.33675)$. The sunset-endpoint note estimated the endpoint coupling by a linear-response slope and used $M'(\hat{r}) = -J(0)$, where the correct relation is $M'(\hat{r}) = -J(0)/2$ (useries-triage); the resulting $\times 1.34$ estimate was invalidated. Here the constant is evaluated DIRECTLY by quadrature, so the linearisation – and the factor-2 error with it – does not enter. The scope is the sunset axis only: this note does not close SC-SCOPE.

2. The M-ENDPOINT constant

In the production convention $\hat{r}(I) = r_R + 2\lambda' I$ with $r_R = \text{gap point at } \mu^2 = 0.005$, $M_R = M(r_R)$, $\lambda' = 3u + 30vM_R$, the dressing variance is the radial integral

$$M(m) = \frac{1}{2\pi^2} \int_0^\infty \frac{k^2}{m + C(k^2 - q_0^2)^2} dk.$$

Two independent quadratures (the production **M_fast** and a direct radial trapezoid) agree to 0.20%:

$$\boxed{M\text{-ENDPOINT} = M(0.33675) = 0.104953} \quad (M_R = 0.109414, \hat{r}(2 \times 10^{-3}) = 0.33675).$$

The value sits below M_R (the dressing at the higher endpoint mass lowers the variance, as monotonicity requires) and inside the corrected convex/secant bracket $[0.10474, 0.10941]$ (the old $[0.1001, 0.1094]$ used the factor-2 slope). $M(0.32064) = 0.107098$ is the $I = 10^{-3}$ companion.

3. The corrected sunset-axis endpoint sign

With $u_{\text{eff}}(M) = u + 10vM$, the matched single- J_0 sunset bound and the composed AddE margin are

$$b(I) = \frac{1}{2} u_{\text{eff}}(M)^2 (2I) \cdot 6J_0 M, \quad \text{composed}(I) = \text{MARGIN} (1 - 1/\rho(I)),$$

with $J_0 = 0.2896$, $\text{MARGIN} = 0.00432$, $\rho(2 \times 10^{-3}) = 2.6$. The frozen-coupling bound (U4) freezes u_{eff} at M_R ; the physically correct endpoint dresses each competitor at its OWN kernel $M(\hat{r}(I))$:

$$\frac{\text{composed}}{b} \Big|_{I=2 \times 10^{-3}} : \quad \underbrace{\times 0.970}_{\text{frozen } M_R (=U4)} \longrightarrow \boxed{\underbrace{\times 1.13}_{\text{dressed } M\text{-ENDPOINT}}} > 1.$$

At $I = 10^{-3}$ the dressed ratio is $\times 3.02$. Thus, with the directly-evaluated M-ENDPOINT, the sunset-axis endpoint is POSITIVE: the U4 marginal failure ($\times 0.97$) was a frozen-coupling artefact, not a real third-cumulant obstruction. This supersedes the invalidated $\times 1.34$ linear estimate; the corrected dressed value is $\times 1.13$.

4. What this resolves and what remains

RESOLVED: M-ENDPOINT (the named missing constant) and, with it, the sunset axis of the third-cumulant endpoint, which is positive at all certified intensities. OPEN (SC-SCOPE remains open): the quartic-difference channel (GHAT4-PERTRANSFER – at sup-kernel grade $\times 1.0$, per-transfer form factor load-bearing) and the tadpole alignment (R-U6-1). The all-orders lift requires these re-assembled as a joint second+third-order inequality; this note advances one of the three load-bearing channels.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*The single- J_0 convention hides a kernel dependence.*” VALID-with-mitigation: J_0 is taken at the anchor dressing to match the certified U4/U7 tables (which the script reproduces: frozen endpoint $\times 0.970$ vs U4 $\times 0.97$). Using J at each $\hat{r}(I)$ would only RAISE the dressed ratios (larger $\hat{r}$ lowers J), so the positive verdict is conservative under this choice.
- (β) “ *$\times 1.13$ is barely above 1 – is the margin real?*” VALID-with-mitigation: the endpoint is the thinnest corner by construction; the sign is robust (M-ENDPOINT cross-checked to 0.20%, J_0 to 0.1%), and the next intensity is already $\times 3.0$. The note claims only the SIGN, not a precise margin.
- (γ) “*Resolving the sunset axis closes SC-SCOPE.*” DISMISSED: explicitly NOT claimed. SC-SCOPE stays OPEN on GHAT4-PERTRANSFER and R-U6-1; this note resolves one of three channels.

Quantitative sanity check (mandatory): dimensional – M has the dimension of M_R and the two quadratures agree to 0.20%; limit-case – the frozen- M_R ratio reproduces the certified U4 $\times 0.97$ to $< 1\%$; sign-direction – $M(\hat{r}) < M_R$ for $\hat{r} > r_R$ (monotonicity), and the dressed coupling is smaller, lowering the sunset bound, raising the ratio, all in the selection-protecting direction.

6. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / SC-SCOPE sunset axis (M-ENDPOINT)
Precise statement:	M-ENDPOINT = $M(0.33675) = 0.104953$ (two quadratures agree 0.20%). With this directly-dressed coupling the sunset-axis endpoint ratio is $\times 1.13 > 1$ (vs frozen-coupling $\times 0.97 = \text{U4}$); $I=1e-3$ gives $\times 3.02$. The sunset axis of the third-cumulant endpoint is POSITIVE.
Scope:	Sunset axis only; matched single- J_0 convention; second + third cumulant sunset channel at the certified intensities. Does NOT close SC-SCOPE.
Dependencies:	production convention (gap_solve, M_fast); J_0 at the anchor dressing; MARGIN=0.00432; AddE rho ratios.

Evidence grade: EXECUTED (scscope_mendpoint_eval.py, 11/11 asserts) + independent-quadrature cross-check.

Reproduction command: python codes/vacuum/scscope_mendpoint_eval.py

Expected output: 11/11 PASS; M-ENDPOINT=0.10495; dressed endpoint x1.13.

Falsification gate: M-ENDPOINT outside [0.1047,0.1094]; the dressed endpoint ratio ≤ 1 (would reinstate the sunset obstruction).

Tier before / after: No tier change. M-ENDPOINT gate: value RESOLVED (recommend RESOLVED pending operator sign-off). SC-SCOPE stays OPEN (GHAT4-PERTRANSFER + R-U6-1).

No-overclaim statement: This resolves ONE of three SC-SCOPE channels. It does NOT close SC-SCOPE, is second/third-cumulant sunset scope only, and the sign (not a precise margin) is the claim. $M' = -J(0)/2$ factor-2 is bypassed by direct quadrature.

Next required action: GHAT4-PERTRANSFER (per-transfer quartic form factor); R-U6-1 (tadpole alignment); then the joint second+third-order endpoint inequality.